Optimization I
Unconstrained Optimization

Quang-Thanh Tran

National Economics University
Department of Mathematical Economics

May 15, 2025

Content

1. Single variable
2. Multiple variables
3. Two variables

FOC, SOC

Single Variable

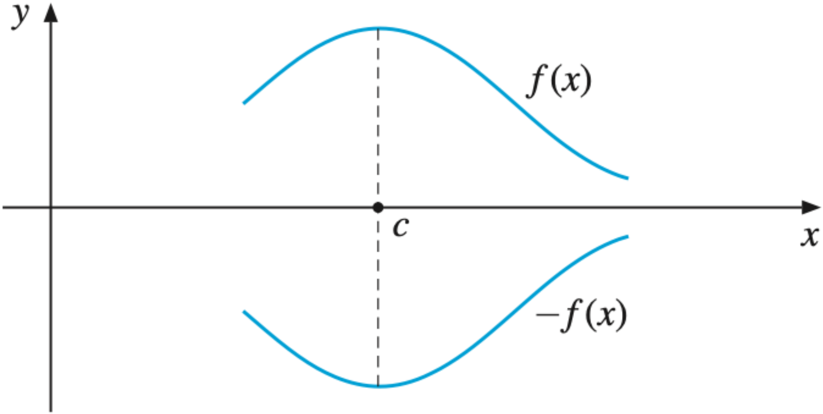
Assumptions

- Objective function: function to be optimized $f(x) \in \mathbb{R}^1$
- Domain: $D \subset \mathbb{R}^1$

Statement

- c is a maximum point of f if and only if $f(c) \geq f(x)$ for all $x \in D$.
- d is a minimum point of f if and only if $f(d) \leq f(x)$ for all $x \in D$.

The maximum point of $f(x)$ is the minimum point of $-f(x)$ (and vice versa).



First-Order Necessary Condition

If c is a max point, what properties does it have?
Assumptions

- f differentiable on the interval $I = [a, b]$
- c is an interior point of I (i.e., $c \in (a, b)$).

If c is a maximum/minimum point of f on I, it must be a critical point of f. Critical points satisfy the first-order necessary condition as follows:

$$f'(c) = 0$$

Proof
Suppose c is an interior point and c is a maximum point. Prove $f'(c) = 0$.
Hint: Suppose there exists an arbitrary sufficiently small number h such that $c + h \in I$. Use the definition of derivative to show $f'(c) = 0$.

▸ Proof

Example

Find critical points of

1. $f(x) = 3 - (x-2)^2$
2. $g(x) = \sqrt{x-5} - 100,\ x \geq 5$

FONC is necessary but not sufficient

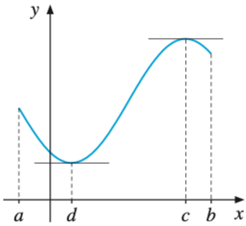


Figure 8.1.2 Two critical points

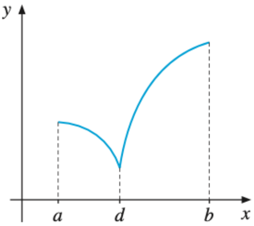


Figure 8.1.3 No critical points

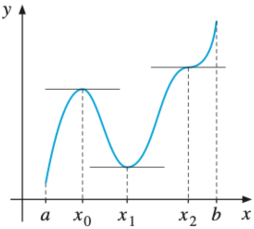


Figure 8.1.4 No interior extrema

How to determine whether a critical point is a maximum or minimum?

Sufficient Conditions

Taylor expansion for a single-variable function. Given a point c and $f(c)$, the value of $f(x)$ for a point x near c is:

$$f(x) \approx f(c) + f'(c)(x-c) + \frac{1}{2}f''(c)(x-c)^2 + \cdots + \frac{1}{n!}f^{(n)}(a)(x-a)^n + R_n$$

where $R_n \to 0$ as $n \to \infty$.
In the problem, treating $O(3)$ as negligible, we can write:

$$f(x) \approx f(c) + f'(c)(x-c) + \frac{1}{2}f''(c)(x-c)^2 + O(3)$$

With $f'(c) = 0$ since c is a critical point, we examine the sign of $f''(c)$.

- If $f''(c) > 0$, then $f(x) \geq f(c) \quad \forall x$ near c, so c is a min
- If $f''(c) < 0$, then $f(x) \leq f(c) \quad \forall x$ near c, so c is a max.

If the sign is strict, this condition is sufficient. If not, we must consider terms up to $O(3)$.

Necessary vs. Sufficient Conditions

To illustrate necessary and sufficient conditions, consider the critical points of x^4.
The critical point is $x = 0$, satisfying $f'(x) = 4x^3 = 0$.
However, $f''(0) = 0$, so it could be either a min or max.
Next, we consider $f'''(0) = 0$, which also cannot determine min or max.
The fourth derivative is $f''''(0) = 24 > 0$.
Now we can determine this is a min point.
If at the second derivative we already had $f''(x^*) > 0$, then this condition is sufficient to determine it is a min point.

Example

Find critical points and determine whether they are min or max.

$$
\begin{aligned}
&(a)\ f(x) = \frac{8}{3x^2 + 4}\\
&(b)\ f(x) = \ln x - 5x\ (x > 0),\\
&(c)\ f(x) = e^x + e^{-2x},\\
&(d)\ f(x) = 9 - (x-a)^2 - 2(x-b)^2
\end{aligned}
$$

Example: Cost Minimization [EMEA:Ex8.2.10]

[Van Dantzig, D. (1956). *Economic decision problems for flood prevention*. Econometrica 276-287.]

After the 1953 North Sea flood disaster, the Dutch government initiated a project to determine the optimal height of dikes. They needed to find the value x such that the cost function

$$C(x) = I_0 + kx + Ae^{-\alpha x}$$

is minimized, where x is the dike height, $I_0 + kx$ is construction cost, and $Ae^{-\alpha x}$ is the estimated flood damage. Note that constants $I_0, k, A, \alpha > 0$ and $A\alpha > k$. Find the optimal value x^* that minimizes $C(x)$.

Example: Profit Maximization [EMEA:Eg8.3.3]

Suppose we have an inverse demand function $p(q)$ where p is price and q is quantity. The average cost is k per unit.
The profit function is

$$\pi(q) = qp(q) - kq$$

1. Find the expression for optimal quantity q^*.
2. How does the optimal quantity change if k changes?
3. How does the optimal profit change if k changes?

Example: Profit Maximization and Subsidies [EMEA:Ex8.3.2]

Suppose

$$p(q) = a - q, \text{ and } 0 < k < a$$

with profit

$$\pi(q) = qp(q) - kq$$

1. Find the optimal quantity and profit $q^*, \pi(q^*)$.
2. How do quantity and profit change with k?
3. The government believes the monopolist produces too little. The government wants the monopolist to produce $\hat{q} = a - k$ units by subsidizing s per unit. Compute the subsidy s required to achieve this goal.

Multiple Variables

Necessary Conditions

Suppose we have

- $\mathbf{x}(x_1, \ldots, x_n)$ is a point in the domain $U \subset \mathbb{R}^n$
- Objective function: $f : U \mapsto \mathbb{R}^1$

If $\mathbf{x}^*$ is a max/min point of f and if $\mathbf{x}^*$ is an interior point of U, then

$$\frac{\partial f}{\partial x_i}(\mathbf{x}^*) = 0 \quad \text{for } i = 1, \ldots, n$$

In other words, its gradient is

$$\nabla f(\mathbf{x}^*) = 0.$$

▸ Proof

To prove SOC, we will again use the Taylor series.

Taylor expansion of $f(x)$ around x^* is

$$f(x) = f(x^*) + \frac{df}{dx}\Big|_{x^*}(x - x^*) + \frac{1}{2}\frac{d^2 f}{dx^2}\Big|_{x^*}(x - x^*)^2 + O(3)$$

Let the perturbation be $\triangle x^* \equiv x - x^*$, rewrite the Taylor expansion as

$$f(x) = \underbrace{f(x^*)}_{zeroth\ \ order} + \underbrace{f'(x^*)\triangle x^*}_{first\ \ order} + \underbrace{\frac{1}{2}f''(x^*)(\triangle x^*)^2}_{second\ \ order} + \underbrace{O(3)}_{higher\ \ order}$$

Assume $O(3)$ is negligible

- min (local): Meaning $f(x^*) \leq f(x)$, FOC is equivalent to first-order term $=$ 0.
- At the same time, if second-order term is positive, then $f(x^*) < f(x)$ (this is the sufficient condition for a min).
- The max problem is similar and opposite.

Taylor Expansion for 2 variables

Taylor expansion of $f(x, y)$ around the optimal point (x^*, y^*) is:

$$
\begin{aligned}
f(x,y) &= f(x^*,y^*) + f_x(x^*,y^*)\triangle x^* + f_y(x^*,y^*)\triangle y^* \\
&+ \frac{1}{2}\left[f_{xx}(x^*,y^*)(\triangle x^*)^2 + 2f_{xy}(x^*,y^*)\triangle x^* \triangle y^* + f_{yy}(x^*,y^*)(\triangle y^*)^2\right] \\
&+ O(3)
\end{aligned}
$$

Written in matrix form

$$
\begin{aligned}
f(x,y) &= f(x^*,y^*) + \begin{bmatrix} f_x(x^*,y^*) & f_y(x^*,y^*) \end{bmatrix} \begin{bmatrix} \triangle x^* \\ \triangle y^* \end{bmatrix} \\
&+ \frac{1}{2}\begin{bmatrix} \triangle x^* & \triangle y^* \end{bmatrix} \begin{bmatrix} f_{xx}(x^*,y^*) & f_{xy}(x^*,y^*) \\ f_{xy}(x^*,y^*) & f_{yy}(x^*,y^*) \end{bmatrix} \begin{bmatrix} \triangle x^* \\ \triangle y^* \end{bmatrix} \\
&+ O(3)
\end{aligned}
$$

Taylor Expansion for 2 variables

(continued)

$$
\begin{aligned}
f(x,y) = {} & f(x^*,y^*) \\
& + \nabla f(x^*,y^*)^T \begin{bmatrix} \triangle x^* \\ \triangle y^* \end{bmatrix} \\
& + \frac{1}{2} \begin{bmatrix} \triangle x^* & \triangle y^* \end{bmatrix} \mathbf{H}(x^*,y^*) \begin{bmatrix} \triangle x^* \\ \triangle y^* \end{bmatrix} \\
& + O(3)
\end{aligned}
$$

where the gradient is:

$$
\nabla f(x^*,y^*) = \begin{bmatrix} f_x(x^*,y^*) \\ f_y(x^*,y^*) \end{bmatrix}
$$

and the Hessian matrix

$$
\mathbf{H}(x^*,y^*) = \begin{bmatrix} f_{xx}(x^*,y^*) & f_{xy}(x^*,y^*) \\ f_{xy}(x^*,y^*) & f_{yy}(x^*,y^*) \end{bmatrix}
$$

Conditions

FOC remains:

$$\nabla f(x^*, y^*) = \begin{bmatrix} f_x(x^*, y^*) \\ f_y(x^*, y^*) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Second-Order Sufficient Conditions

- min (local): red-colored term > 0 for any $(\triangle x^*, \triangle y^*)$, equivalent to matrix $\mathbf{H}$ being positive definite
- max (local): red-colored term < 0 for any $(\triangle x^*, \triangle y^*)$, equivalent to matrix $\mathbf{H}$ being negative definite.

For example, if this Hessian matrix is

$$\mathbf{H} = \begin{bmatrix} 4 & -2 \\ -2 & 3 \end{bmatrix}$$

Prove that $\frac{1}{2} v^T \mathbf{H} v$ is positive for any $v = (\Delta x, \Delta y)$.

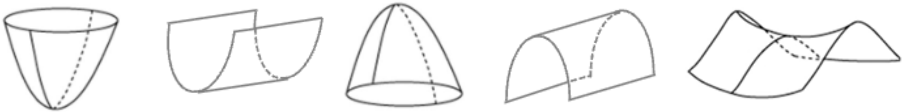
Proof

Definiteness

Consider the critical point (x^*, y^*) and objective function $z = f(x^*, y^*)$. Let $\mathbf{v} = (\Delta x, \Delta y)$, we have:

definiteness	definition	max/min	shape
positive definite **H**	$\mathbf{v}^T\mathbf{H}\mathbf{v} > 0$	min	inverted cone
positive semidefinite **H**	$\mathbf{v}^T\mathbf{H}\mathbf{v} \geq 0$	inconclusive	valley
negative definite **H**	$\mathbf{v}^T\mathbf{H}\mathbf{v} < 0$	max	cone (mountain peak)
negative semidefinite **H**	$\mathbf{v}^T\mathbf{H}\mathbf{v} \leq 0$	inconclusive	tunnel through mountain
indefinite **H**	$\mathbf{v}^T\mathbf{H}\mathbf{v} \lesseqgtr 0$	neither	saddle point

Note that only positive definite/negative definite are sufficient conditions. If the Hessian is positive semidefinite/negative semidefinite, these are only necessary but not sufficient.



Taylor Expansion for n variables

Similarly, we obtain

$$
\begin{aligned}
f(\mathbf{x}) = {} & f(\mathbf{x}^*) + \sum_{i=1}^{n} \frac{\partial f(\mathbf{x}^*)}{\partial x_i}(x_i - x_i^*) \\
& + \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \frac{\partial^2 f(\mathbf{x}^*)}{\partial x_i \partial x_j}(x_i - x_i^*)(x_j - x_j^*) + O(3) \\
& = f(\mathbf{x}^*) + \nabla f^T(\mathbf{x}^*) \triangle \mathbf{x}^* + \frac{1}{2} \triangle \mathbf{x}^{*T} \mathbf{H}(\mathbf{x}^*) \triangle \mathbf{x}^* + O(3)
\end{aligned}
$$

FOC

$$\nabla f(\mathbf{x}^*) = \begin{bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \\ \vdots \\ \partial f/\partial x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Hessian matrix

$$\mathbf{H}(\mathbf{x}^*) = \begin{pmatrix} \dfrac{\partial^2 f}{\partial x_1^2}(\mathbf{x}^*) & \dots & \dfrac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \dfrac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}^*) & \dots & \dfrac{\partial^2 f}{\partial x_n^2}(\mathbf{x}^*) \end{pmatrix}$$

SOC:

- If $\mathbf{H}$ is negative definite, $\mathbf{x}^*$ is a max (local)
- If $\mathbf{H}$ is positive definite, $\mathbf{x}^*$ is a min (local)
- If neither of the two cases above, $\mathbf{x}^*$ is a saddle point.

Testing Definiteness of the Hessian

Using Sylvester's criterion

We use leading principal minors.

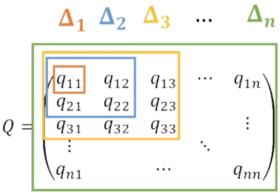


Figure: n-leading principal minors

▷ If the determinant of the n leading principal minors of $\mathbf{H}(\mathbf{x}^*)$ is positive, i.e.:

$$|f_{x_1x_1}| > 0, \begin{vmatrix} f_{x_1x_1} & f_{x_2x_1} \\ f_{x_1x_2} & f_{x_2x_2} \end{vmatrix} > 0, \begin{vmatrix} f_{x_1x_1} & f_{x_2x_1} & f_{x_3x_1} \\ f_{x_1x_2} & f_{x_2x_2} & f_{x_3x_2} \\ f_{x_1x_3} & f_{x_2x_3} & f_{x_3x_3} \end{vmatrix} > 0, \ldots$$

then $\mathbf{H}(\mathbf{x}^*)$ is positive definite and $\mathbf{x}^*$ is a min (local) of f.

Testing Definiteness of the Hessian

▷ If the determinant of the n leading principal minors of $\mathbf{H}(\mathbf{x}^*)$ alternates in sign continuously:

$$\left|f_{x_1x_1}\right| < 0, \begin{vmatrix} f_{x_1x_1} & f_{x_2x_1} \\ f_{x_1x_2} & f_{x_2x_2} \end{vmatrix} > 0, \begin{vmatrix} f_{x_1x_1} & f_{x_2x_1} & f_{x_3x_1} \\ f_{x_1x_2} & f_{x_2x_2} & f_{x_3x_2} \\ f_{x_1x_3} & f_{x_2x_3} & f_{x_3x_3} \end{vmatrix} < 0, \ldots$$

then $\mathbf{H}(\mathbf{x}^*)$ is negative definite and $\mathbf{x}^*$ is a max of f.

▷ If the signs of the principal minors do not follow the above rule, then $\mathbf{x}^*$ is a saddle point of f. It is neither a max nor a min. It is a max along the direction of one variable but a min along the direction of another variable.

References:

- Notes in Numerical Analysis
- Notes in Applied Mathematics
- Cholesky decomposition of positive definite matrices

Given the objective function $F(x, y) : U \mapsto \mathbb{R}^1$ twice differentiable with domain $U \subset \mathbb{R}^2$. Optimality conditions

- FOC

$$f_x(x^*, y^*) = f_y(x^*, y^*) = 0$$

- SOC

$$\mathbf{H}(x^*, y^*) = \begin{bmatrix} f_{xx}(x^*, y^*) & f_{yx}(x^*, y^*) \\ f_{xy}(x^*, y^*) & f_{yy}(x^*, y^*) \end{bmatrix}$$

  - If $f_{xx} < 0, \det(\mathbf{H}) > 0$, then (x^*, y^*) is a local max of F.
  - If $f_{xx} > 0, \det(\mathbf{H}) > 0$, then (x^*, y^*) is a local min of F.
  - If $\det(\mathbf{H}) < 0$, then (x^*, y^*) is a saddle point.

Example

$$\max(\min) f(x,y) = x^3 - y^3 + 9xy$$

FOC

$$(x): \frac{\partial f}{\partial x} = 3x^2 + 9y = 0 \iff y = -x^2/3,$$
$$(y): \frac{\partial f}{\partial y} = -3y^2 + 9x = 0 \iff y^2 = 3x.$$

Critical points

$$(x^*, y^*) = (0,0) \text{ and } (3,-3)$$

The Hessian matrix is

$$H = \begin{pmatrix} 6x & 9 \\ 9 & -6y \end{pmatrix}$$

After checking the determinant signs $|H_1| = 6x$ and $|H_2| = |H|$.

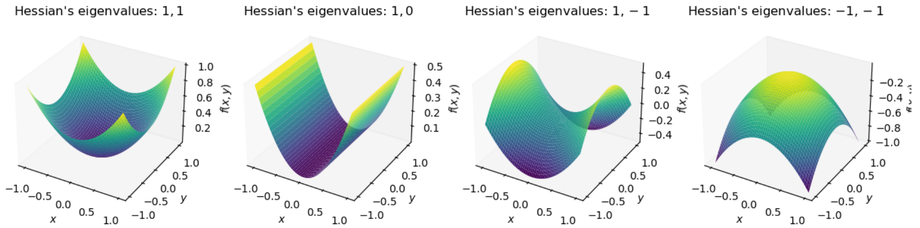
- $(0,0)$ is a saddle point because $|H| = -81 < 0$.
- $(3,-3)$ is a min point because $|H_1| = 18 > 0, |H_2| = 243 > 0$.

Sufficient Conditions: Eigenvalue Test

Theorem (Eigenvalues Test)

If the Hessian at a critical point has

- ***all positive eigenvalues*** $\Rightarrow$ *H is positive-definite* $\Rightarrow$ ***local minimum*** *(convex/concave up)*
- ***all negative eigenvalues*** $\Rightarrow$ *H is negative-definite* $\Rightarrow$ ***local maximum*** *(concave down)*
- ***eigenvalues have one negative, one positive*** $\Rightarrow$ *H is indefinite* $\Rightarrow$ ***saddle***



Finding Eigenvalues

Characteristic equation:

$$\mathbf{Ax} = \lambda \mathbf{x} \quad \Longrightarrow \quad \det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

Example: 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$\begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = (a-\lambda)(d-\lambda) - bc = 0$$

Verification checks:

1. Trace: $\lambda_1 + \lambda_2 = a + d$
2. Determinant: $\lambda_1 \lambda_2 = ad - bc$

Worked example: $B = \begin{pmatrix} 2 & 3 \\ 5 & 4 \end{pmatrix}$ Show that the eigenvalues are $\lambda = \{7, \ -1\}$

Interpreting Eigenvalues of the Hessian

Eigenvalues	Hessian definiteness	Critical point
all > 0	positive definite	local min
all < 0	negative definite	local max
mixed signs	indefinite	saddle point
any $= 0$	singular	inconclusive

Geometric meaning:

- Positive eigenvalues $\Rightarrow$ function curves **up** in those eigenvector directions.
- Negative eigenvalues $\Rightarrow$ function curves **down**.
- Mixed $\Rightarrow$ concave up in one direction, concave down in another $\Rightarrow$ saddle.

Example: $f(x, y) = x^3 + 2y^3 - xy$ at $(0, 0)$

$$\mathbf{H} = \begin{pmatrix} 6x & -1 \\ -1 & 6y \end{pmatrix}_{(0,0)} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

Eigenvalues $\lambda = \pm 1$ (mixed) $\Rightarrow$ saddle point.

1. Find eigenvalues of

$$\text{(a)} \begin{pmatrix} 3 & 8 \\ 4 & -1 \end{pmatrix} \qquad \text{(b)} \begin{pmatrix} 2 & 6 \\ -1 & 3 \end{pmatrix}$$

2. For each Hessian, compute eigenvalues and classify definiteness:

$$\text{(a)} \begin{pmatrix} 12x^2 & -1 \\ -1 & 2 \end{pmatrix} \text{ at } (3,1) \qquad \text{(b)} \begin{pmatrix} 6x & 0 \\ 0 & 6y \end{pmatrix} \text{ at } (-1,2)$$

$$\text{(c)} \begin{pmatrix} -2y^2 & -4xy \\ -4xy & -2x^2 \end{pmatrix} \text{ at } (1,-1) \text{ and } (1,0)$$

3. Determine concavity of $f(x,y) = x^3 + 2y^3 - xy$ at $(0,0)$, $(3,3)$, $(3,-3)$, $(-3,3)$, $(-3,-3)$.
4. Find critical points and use eigenvalues to classify:

$$\text{(a) } f(x,y) = 4x + 2y - x^2 - 3y^2 \qquad \text{(b) } f(x,y) = x^4 + y^2 - xy$$

Applications

Practice

Find critical points, check whether they are min, max, or saddle.

1. $f(x, y) = 4x^2 - xy + y^2 - x^3$
2. $f(x, y) = 3x^4 + 3x^2y - y^3$
3. $f(x, y, z) = 2x^2 + y^2 + 4z^2 - x + 2z$
4. $f(x, y, z) = x^2 + 6xy + y^2 - 3yz + 4z^2 - 10x - 5y - 21z$

Price Discrimination [EMEA:Eg13.4.1]

The monopolist sells a good in 2 markets with different demands

$$p_1 = a_1 - b_1 q_1,$$
$$p_2 = a_2 - b_2 q_2$$

The total cost is

$$C(Q) = \alpha Q = \alpha(q_1 + q_2)$$

1. Write the profit function.
2. Find the optimal (q_1^*, q_2^*) that maximizes the monopolist's profit.
3. What prices will the monopolist charge in each market?
4. Find the maximized profit.
5. What if price discrimination is not allowed? Compute the loss given $a_1 = 100, a_2 = 80, b_1 = b_2 = 1$.

Duopoly [EMEA:Ex13.4.5]

Two firms, A and B, produce the same products with quantities x and y, sold at prices p and q.
Demands for two brands are

$$\begin{aligned} x &= 29 - 5p + 4q, \\ y &= 16 + 4p - 6q \end{aligned}$$

Cost functions to each firm

$$\begin{aligned} c(x) &= 5 + x, \\ d(y) &= 3 + 2y \end{aligned}$$

1. Two firms collude to maximize their combined profit. Find the optimal prices (p, q) and production levels (x, y).
2. An antitrust authority then prohibits collusion. Find the optimal production levels of each firm, taking the other's price as given.
3. What are the equilibrium prices?

Given an i.i.d sample $\{x_1, \ldots, x_n\}$ that follows a normal distribution $N(\mu, \sigma^2)$. Estimate μ, σ^2 using maximum likelihood.

1. Write the pdf function

$$f(x_i|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

2. Write the likelihood function

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} f(x_i|\mu, \sigma^2)$$

3. Take logs to derive the log likelihood function $\ell(\mu, \sigma^2)$.
4. Derive the first-order conditions of $\hat{\mu}, \hat{\sigma}^2$ to maximize $\ell(\mu, \sigma^2)$.

OLS [Reference]

Suppose the regression function at observation i is

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \tag{1}$$

We need to determine β_0, β_1 such that the error ϵ is minimized. With n observations, we have the problem

$$\min_{\hat{\beta}_0, \hat{\beta}_1} W = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \tag{2}$$

1. Write the FOC, knowing that $\sum_{i=1}^{n} y_i \equiv n\overline{y}$ and $\sum_{i=1}^{n} x_i \equiv n\overline{x}$.
2. Prove that $\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$.
3. Using the above result, prove the blue part

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} x_i y_i - n\overline{xy}}{\sum_{i=1}^{n} x_i^2 - n\overline{x}^2} \stackrel{?}{=} \frac{\sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n} (x_i - \overline{x})^2}$$

4. Prove $\sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y}) = \sum_{i=1}^{n} x_i y_i - n\overline{xy}$ and $\sum_{i=1}^{n} (x_i - \overline{x})^2 = \sum_{i=1}^{n} x_i^2 - n\overline{x}^2$. Hence show that $\hat{\beta}_1$ can be rewritten as the red part.

Appendix

References

EMEA : Essential mathematics for economic analysis (Sysaeter & Hammond et. al, 6e 2021)

FMMA : Fundamental methods of mathematical economics (Chiang & Wainwright, 4e 2005)

FMEA : Further mathematics for economic analysis (Sysaeter & Hammond et. al, 2e 2008)

ME : Mathematics for economists (Simon & Blume, 1994)

NMO : Numerical methods and optimization (Butenko & Pardalos, 2014)

ORAA : Operations Research: Applications and Algorithms (Winston, 2003)

Proof 1

Suppose $c \in (a, b)$ is a maximum point of f on the interval $I = [a, b]$. By definition of a maximum point, there exists a neighborhood of c such that for all x sufficiently close to c, we have $f(x) \leq f(c)$. Take a sufficiently small number h such that $c + h \in I$.

Proof

Case $h > 0$**:**

Since c is a maximum point, $f(c+h) \leq f(c)$, implying $\frac{f(c+h)-f(c)}{h} \leq 0$.
Taking the limit as $h \to 0^+$, we obtain

$$\lim_{h \to 0^+} \frac{f(c+h) - f(c)}{h} \leq 0.$$

Case $h < 0$**:**

Similarly, from $f(c+h) \leq f(c)$ we have $\frac{f(c+h)-f(c)}{h} \geq 0$. Taking the limit as $h \to 0^-$, we obtain

$$\lim_{h \to 0^-} \frac{f(c+h) - f(c)}{h} \geq 0.$$

Combining the two inequalities, we obtain

$$f'(c) \equiv \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = 0$$

Proof 2

Consider the vector $v = (\Delta x, \Delta y)^T$ and matrix

$$\mathbf{H} = \begin{bmatrix} 4 & -2 \\ -2 & 3 \end{bmatrix}.$$

Consider:

$$\frac{1}{2} v^T \mathbf{H} v.$$

First compute $\mathbf{H}v$

$$\mathbf{H}v = \begin{bmatrix} 4 & -2 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} 4\Delta x - 2\Delta y \\ -2\Delta x + 3\Delta y \end{bmatrix}.$$

Thus:

$$\begin{aligned} v^T \mathbf{H} v &= \begin{bmatrix} \Delta x & \Delta y \end{bmatrix} \begin{bmatrix} 4\Delta x - 2\Delta y \\ -2\Delta x + 3\Delta y \end{bmatrix} \\ &= \Delta x(4\Delta x - 2\Delta y) + \Delta y(-2\Delta x + 3\Delta y) \\ &= 4\Delta x^2 - 2\Delta x\Delta y - 2\Delta x\Delta y + 3\Delta y^2 \\ &= 4\Delta x^2 - 4\Delta x\Delta y + 3\Delta y^2. \end{aligned}$$

Then

$$\frac{1}{2}v^T\mathbf{H}v = 2\Delta x^2 - 2\Delta x\Delta y + \frac{3}{2}\Delta y^2.$$

We rewrite the above expression as a sum of squares

$$2\Delta x^2 - 2\Delta x\Delta y + \frac{3}{2}\Delta y^2 = 2\left(\Delta x - \frac{1}{2}\Delta y\right)^2 + \Delta y^2.$$

Since:

$$2\left(\Delta x - \frac{1}{2}\Delta y\right)^2 \geq 0 \forall \Delta x, \Delta y, \quad \Delta y^2 \geq 0.$$

These two terms are both zero only when

$$\Delta x = \Delta y = 0.$$

Therefore, for all $v \neq 0$, we have

$$\frac{1}{2}v^T\mathbf{H}v > 0.$$

Thus $\frac{1}{2}v^T\mathbf{H}v$ is always positive for all $v = (\Delta x, \Delta y) \neq (0, 0)$.

General FONC for Unconstrained Optimization

Statement: Let $U \subset \mathbb{R}^n$ be open, $f : U \to \mathbb{R}$ differentiable. If $\mathbf{x}^* \in U$ is a local extremum of f, then

$$\nabla f(\mathbf{x}^*) = \mathbf{0} \quad \text{i.e.} \quad \frac{\partial f}{\partial x_i}(\mathbf{x}^*) = 0, \ \forall i.$$

Proof:

1. Choose an arbitrary vector $\mathbf{v} \in \mathbb{R}^n$. Consider the single-variable function
$$g(t) = f(\mathbf{x}^* + t\mathbf{v}), \quad t \in \mathbb{R}.$$
Since U is open, there exists $\delta > 0$ such that $g(t)$ is defined for $|t| < \delta$.
2. Since $\mathbf{x}^*$ is a local extremum of f, there exists $V \subset U$ of x^* such that $f(\mathbf{x}) \leq f(\mathbf{x}^*)$ (if maximum) or $f(\mathbf{x}) \geq f(\mathbf{x}^*)$ (if minimum) for all $\mathbf{x} \in V$. Then, for sufficiently small t, $\mathbf{x}^* + t\mathbf{v} \in V$, implying $g(t) \leq g(0)$ (maximum) or $g(t) \geq g(0)$ (minimum). Hence, $t = 0$ is a local extremum of g. $t = 0$ is a local extremum of g. For a single-variable function, the necessary condition for an extremum is $g'(0) = 0$.
3. Next, we prove $g'(0) = 0$ using the chain rule:
$$g'(t) = \nabla f(\mathbf{x}^* + t\mathbf{v}) \cdot \mathbf{v} \quad \Rightarrow \quad g'(0) = \nabla f(\mathbf{x}^*) \cdot \mathbf{v}.$$
At $t = 0$:
$$g'(0) = \nabla f(\mathbf{x}^*) \cdot \mathbf{v} = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{x}^*) v_i$$
4. Since $\mathbf{v}$ is an arbitrary vector in $\mathbb{R}^n$, $\nabla f(\mathbf{x}^*) \cdot \mathbf{v} = 0$ for all $\mathbf{v} \in \mathbb{R}^n$. This only occurs when $\nabla f(\mathbf{x}^*) = \mathbf{0}$.

Note: This condition only holds when $\mathbf{x}^*$ is an interior point of the domain. If $\mathbf{x}^*$ lies on the boundary, the Kuhn–Tucker conditions must be considered.